Hybrid Video Provenance Framework (HVPF)

Distinguishing Organic, Manipulated and Synthetic Sensor Noise in Digital Videos
Implementation Progress Report

Talha Aslam
MS Thesis

5 October 2026

Abstract

This report documents the implementation work carried out so far for the Hybrid Video Provenance Framework (HVPF) proposed in the thesis. A complete, reproducible nine-step pipeline has been built: dataset inventory, leakage-free splits, I-frame extraction, PRNU fingerprinting with a conventional PCE baseline, generation of a manipulated class (PRNU removal, PRNU injection and denoising attacks), an AI-generated (synthetic) class, feature extraction (PRNU, residual statistics, FFT, temporal sensor-noise consistency (TSNCS) and Noiseprint++), and classical, deep temporal and Transformer classifiers with the evaluation protocol of the proposal. Steps 1–6 are complete and verified; Steps 7–9 are implemented and tested end to end, and the full feature extraction and model training are running. The PRNU baseline reproduces the expected behaviour on VISION (100% attribution of native videos from non-stabilised devices), and several findings relevant to RQ1 and RQ2 have already emerged. The code is available at `https://github.com/TalhaAslam44/PRNU-Research`.

Contents

1 Executive summary **3**

2 Objectives and research questions **3**

3 System overview **4**

4 Datasets **4**
4.1 Inventory and quality control (Step 1) 5

5 Experimental protocol (Step 2) **5**

6 Frame extraction (Step 3) **6**

7 PRNU fingerprints and conventional baseline (Step 4) **6**
7.1 Method 6
7.2 VISION results 6
7.3 Rotation finding (RQ2) 7
7.4 Video-ACID results 7

8 Manipulated class (Step 5) **8**
8.1 Attack design 8
8.2 Verification 9

9 Synthetic class (Step 6) **9**

10 Feature extraction (Step 7) **10**

11 Models and evaluation (Steps 8–9) **10**

12 Deviations from the proposal **11**

13 Technical issues and resolutions **12**

14 Current status and next steps **12**

A VISION devices **13**

1 Executive summary

Completed.

- **Data:** VISION (35 devices, 1,914 videos in native, YouTube and WhatsApp versions) and Video-ACID (46 devices, 1,840 videos streamed from 177 GB of archives) inventoried and split without leakage; 55,959 analysis I-frames extracted.
- **Conventional baseline:** per-device PRNU fingerprints and PCE matching for 79 cameras.
- **Manipulated class:** 4,400 attacked clips (400 videos × 11 variants, 131,879 frames) with removal, injection and denoising attacks at several strengths, including a white-box removal that drives PCE to zero.
- **Synthetic class:** 1,492 AI-generated clips from eight generators (Sora, Kling, CogVideo, OpenSora, CogVideoX, EasyAnimate, HunyuanVideo, LTX-Video).
- **Features and models:** all feature extractors (incl. GPU Noiseprint++), six model families and the full evaluation (PVA, F1, ROC-AUC, EER, ablation) implemented and tested.

Key findings so far.

1. Native videos of non-stabilised VISION devices are attributed with 100% accuracy (AUC 1.000); stabilised devices fall to 32%, confirming that in-camera stabilisation defeats PRNU.
2. On social-media copies, PCE evaluated at zero shift is far stronger than the usual blind peak search (YouTube AUC 0.972 vs 0.812; WhatsApp 0.868 vs 0.713).
3. Social-media copies of *rotated* recordings lose their PRNU almost entirely (WhatsApp AUC 0.52 vs 0.92): WhatsApp's rotation path also rescales the video by 0.5–1% and shifts it by 3–5 pixels. Geometry, not compression, breaks PRNU here (RQ2).
4. Video-ACID is bimodal: DSLRs, action cameras and Samsung phones reach 95–100% attribution, while 19 of 44 devices (mostly phones and compact cameras, likely electronically stabilised) show no usable video PRNU.
5. PRNU removal easily over-subtracts: beyond $\alpha \approx 0.3$–0.5 the PCE becomes strongly negative (e.g. $-4{,}765$ at $\alpha = 1$), itself a detectable trace (RQ1). Injection leaves the true camera's fingerprint next to the victim's.
6. Noiseprint++ does not attribute cameras on luma-only video I-frames (about 20% top-1 on 10 cameras), so it is used for output statistics only.
7. Every clip we re-encode shares an encoder grid that camera originals do not have. An *encoder-controlled* evaluation protocol was added so that this cannot inflate the results.

2 Objectives and research questions

The thesis proposes HVPF, a framework that decides whether the sensor noise in a video is *organic* (produced by a real camera), *manipulated* (suppressed, transferred or injected by an anti-forensic attack) or *synthetic* (the video is AI-generated). The implementation is organised around the three research questions:

RQ1 Which characteristics discriminate organic sensor noise from manipulated or synthetic noise?

RQ2 How do processing operations (compression, resizing, filtering, platform re-encoding) affect PRNU-based attribution and noise-based analysis?

RQ3 Does a hybrid PRNU + machine learning / deep learning framework outperform conventional forensic methods for provenance verification and anti-forensic attack detection?

3 System overview

Figure 1 shows the implemented pipeline. Each step is a separate, resumable script driven by one configuration file (`config.yaml`); finished work is skipped on re-runs.

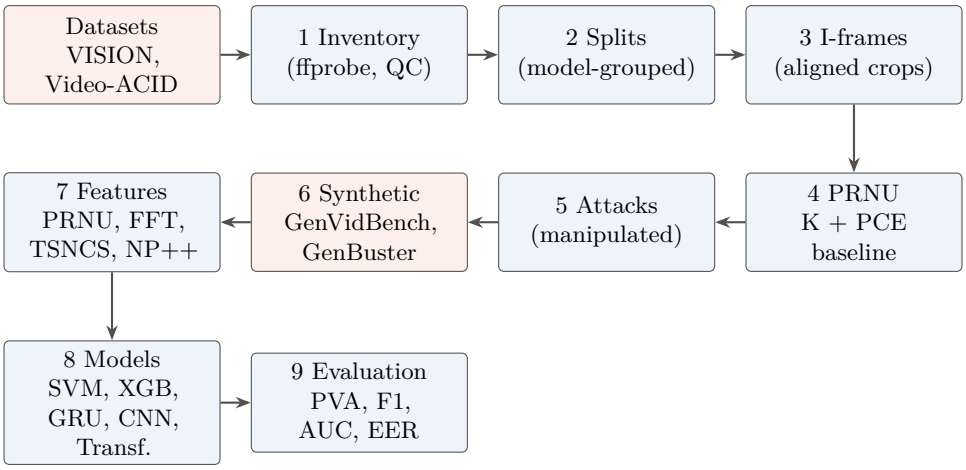


Figure 1: Implemented HVPF pipeline. Orange boxes are data sources; blue boxes are pipeline steps.

Table 1: Implementation environment.

Component	Details
Hardware	16-core CPU, 31 GB RAM, NVIDIA GeForce RTX 3080 (10 GB, CUDA), spinning hard disk
Language and libraries	Python 3.12, PyTorch 2.14 (CUDA), NumPy 2.5, SciPy 1.18, scikit-learn 1.9, XGBoost 3.4, OpenCV, pandas 3.0, PyWavelets
Video tools	FFmpeg / FFprobe 6.1 (libx264 encoder)
PRNU	Politecnico di Milano `prnu-python` [12], patched for NumPy 2
Noiseprint++	GRIP-UNINA TruFor [11] (nonprofit research licence)
Code	`https://github.com/TalhaAslam44/PRNU-Research`

4 Datasets

Table 2: Datasets used in the implementation.

Dataset	Content	Videos	Role
VISION [1]	35 devices, 11 brands; native videos and their YouTube and WhatsApp copies; flat, indoor and outdoor scenes	1,914	organic, attack source
Video-ACID [2]	46 devices of 36 camera models (phones, DSLRs, compacts, camcorders, action cameras); first 40 videos per device	1,840	organic
GenVidBench [4]	AI-generated: Sora, Kling, CogVideo, OpenSora	892	synthetic
GenBuster-200K-mini [5]	AI-generated: CogVideoX, EasyAnimate, HunyuanVideo, LTX-Video	600	unseen-generator test
FloreView [3]	46 smartphones (videos only, about 95 GB)	–	pending

4.1 Inventory and quality control (Step 1)

Every video was probed with FFprobe (codec, resolution, rotation flag, frame rate, duration, number of I-frames) and the full packet stream was read to detect truncated files.

- **VISION:** 1,914 videos (648 native, 622 YouTube, 644 WhatsApp), all H.264. One file is truncated (`D33_V_indoorYT_panrot_0001.mp4`, "moov atom not found") and six videos listed as misplaced in the VISION errata are excluded. About 1 I-frame per second (median 70 per native video); YouTube copies have about 15.
- **Stabilisation:** 14 VISION devices use in-camera digital stabilisation (D02, D05, D06, D10, D12, D14, D15, D18, D19, D20, D25, D29, D32, D34), following the VISION paper and Mandelli et al. [9]. They are kept and reported separately.
- **Video-ACID:** the archives (177 GB) exceed the free disk space, so they are streamed: the first 40 videos of each device are kept, probed and reduced to I-frames, and the videos are deleted. ACID clips are short (median 5–7 s, about 6 I-frames). The Nokia 6.1 (2 devices, 80 videos) uses an MP4 feature FFmpeg 6.1 cannot read and is excluded.
- **Resolution:** two VISION devices record at 800×480 (D04) and 640×480 (D23), which fixed the analysis crop at 480×480 (the proposal suggested 512×512).

The per-device table for VISION (the planned Chapter 5 table) is given in Appendix A.

5 Experimental protocol (Step 2)

Device- and model-disjoint splits. The split unit is the *camera model*: all devices of one model (e.g. the three iPhone 5c units D05, D14, D18) are kept in the same split, and VISION and ACID devices of the same phone model (e.g. Galaxy S5) are grouped across datasets. This is stricter than the device split of the plan, because Noiseprint is a model-level trace. Every version of a recording (native, YouTube, WhatsApp and later the attacked copies) inherits its device's split. Groups are dealt 60/20/20 within strata (dataset mix × stabilisation). Assertions verify that no device, model or recording crosses splits.

Cross-validation. With 35 VISION devices a single test split holds only five cameras, so a 5-fold grouped assignment is also produced; results will be reported as mean ± standard deviation over folds, with the fixed split as a secondary view.

Table 3: VISION devices per split and per fold (stabilised devices in brackets).

Split / fold	Devices
Train (22)	D01 [D02] D03 [D05] D08 D09 [D10] D13 [D14] D17 [D18] D24 [D25] D26 D27 D28 [D29] D30 D31 [D32] [D34] D35
Validation (8)	D04 [D06] D11 [D15] D16 [D20] D22 D33
Test (5)	D07 [D12] [D19] D21 D23
Fold 0	D01 D07 D09 [D20] [D25] D26 D27 D30 [D32]
Fold 1	[D05] D08 D11 D13 [D14] [D18] D28 D33
Fold 2	D03 [D06] [D15] D16 [D19] D23 D35
Fold 3	[D02] D04 [D10] [D12] D22 D24
Fold 4	D17 D21 [D29] D31 [D34]

ACID devices are split in the same way (27 train, 11 validation, 8 test).

Roles. VISION native *flat* videos are *reference* videos that estimate each camera's fingerprint; their YouTube/WhatsApp copies are not used, so the reference never leaks into evaluation. Natural

(indoor/outdoor) videos in all versions are *samples*. ACID has no flat videos: the first 20 videos of each device are references and the next 20 are samples.

6 Frame extraction (Step 3)

Up to 30 I-frames per video (evenly spaced over all I-frames) are decoded, converted to luma and center-cropped to 480×480. Because PRNU is a pixel-wise pattern, frames are kept in the *sensor* orientation: videos are decoded without applying their rotation flag. YouTube and WhatsApp copies bake the rotation into the pixels and are downscaled (e.g. WhatsApp stores 848×480 or 480×848), so they are rotated back and rescaled to the native resolution before cropping. A content check confirmed the alignment (correlation 0.95–0.997 between native and aligned WhatsApp frames). Result: 1,480 VISION videos (39,404 frames, median 30 per video) and 1,760 ACID videos (16,555 frames, median 7).

7 PRNU fingerprints and conventional baseline (Step 4)

7.1 Method

For each camera the maximum-likelihood fingerprint $\hat{K} = \sum_i W_i I_i / \sum_i I_i^2$ [7] is estimated from its reference frames, where W_i is the wavelet-denoised residual of frame I_i. The Politecnico di Milano implementation [12] was used; it needed two fixes (a NumPy 2 incompatibility, and its multi-image estimator only handled RGB), and the grayscale estimator matches the original to 6×10^{-7}. Each sample video gets its own fingerprint from its I-frames, which is compared with every camera of the dataset by:

- **PCE (blind):** peak-to-correlation energy with the peak searched over all shifts [6];
- **PCE$_0$:** PCE evaluated at zero shift, since our frames are aligned by construction;
- **NCC:** normalised correlation at zero shift.

Closed-set attribution picks the camera with the highest score; verification uses the common threshold PCE > 60.

7.2 VISION results

Table 4: PRNU baseline on VISION (35 cameras, closed set). Accuracy = closed-set attribution; AUC and EER for verification of matching vs non-matching camera pairs.

Version	Devices	Queries	Acc. PCE$_0$	AUC PCE$_0$	AUC PCE	EER PCE$_0$	TPR@60
Native	non-stabilised	259	100.0%	1.000	0.999	0.000	0.954
Native	stabilised	166	31.9%	0.738	0.564	0.344	0.145
YouTube	non-stabilised	249	84.7%	0.972	0.812	0.084	0.257
YouTube	stabilised	162	20.4%	0.638	0.489	0.437	0.031
WhatsApp	non-stabilised	257	60.7%	0.868	0.713	0.214	0.257
WhatsApp	stabilised	166	7.8%	0.571	0.464	0.453	0.006

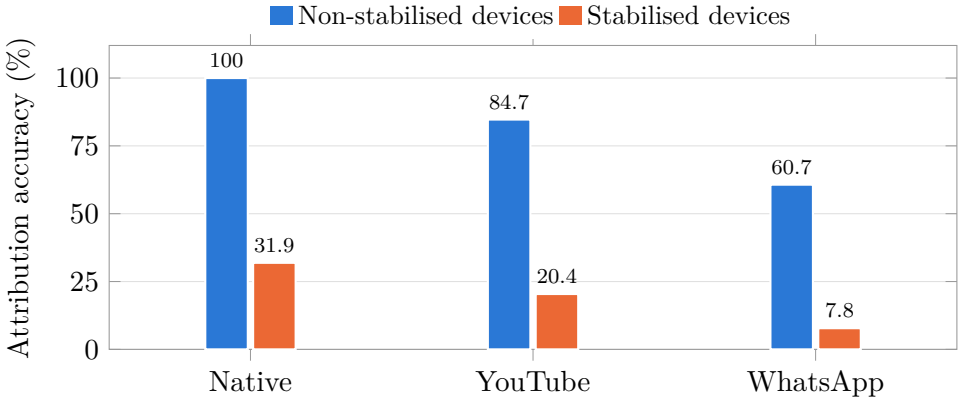


Figure 2: Closed-set attribution (PCE_0) on VISION by version.

Observations. Native videos of non-stabilised devices are attributed perfectly, consistent with published results on VISION [1]; digital stabilisation breaks PRNU, as reported in [9]. On social-media copies PCE_0 is clearly better than the blind peak search, so PCE_0 is used as the conventional method for RQ3. The fixed threshold of 60 is very conservative for social copies (true-positive rate about 0.26 at zero false positives).

7.3 Rotation finding (RQ2)

Breaking the non-stabilised results down by whether the original recording carried a rotation flag reveals a strong effect on social-media copies (Table 5).

Table 5: Non-stabilised VISION devices: effect of the original recording's rotation flag.

Version	Original rotated	Queries	Acc. PCE_0	AUC PCE_0
Native	no	225	100.0%	1.000
Native	yes	34	100.0%	1.000
YouTube	no	222	88.3%	0.982
YouTube	yes	27	55.6%	0.896
WhatsApp	no	223	70.0%	0.920
WhatsApp	yes	34	0.0%	0.523

Native rotated videos are unaffected, so the geometric inversion is correct. Phase correlation between native and WhatsApp frames showed that WhatsApp copies of rotated recordings are offset by 3–5 pixels (versus 0.35 pixels for unrotated ones) and an affine registration found an extra scaling of 0.5–1%. Registering the frames to the native video restored the match (e.g. PCE from 44 to 213). Hence a platform's handling of rotation, not its compression, removes the PRNU in these copies; a blind analyst would need a scale search to recover it.

7.4 Video-ACID results

ACID is much harder: 50.7% closed-set attribution (PCE_0, 44 cameras) and AUC 0.80, because there are no flat videos (fingerprints come from about 120 natural frames per camera) and many 2017–18 devices stabilise electronically. Per device the result is bimodal (Figure 3).

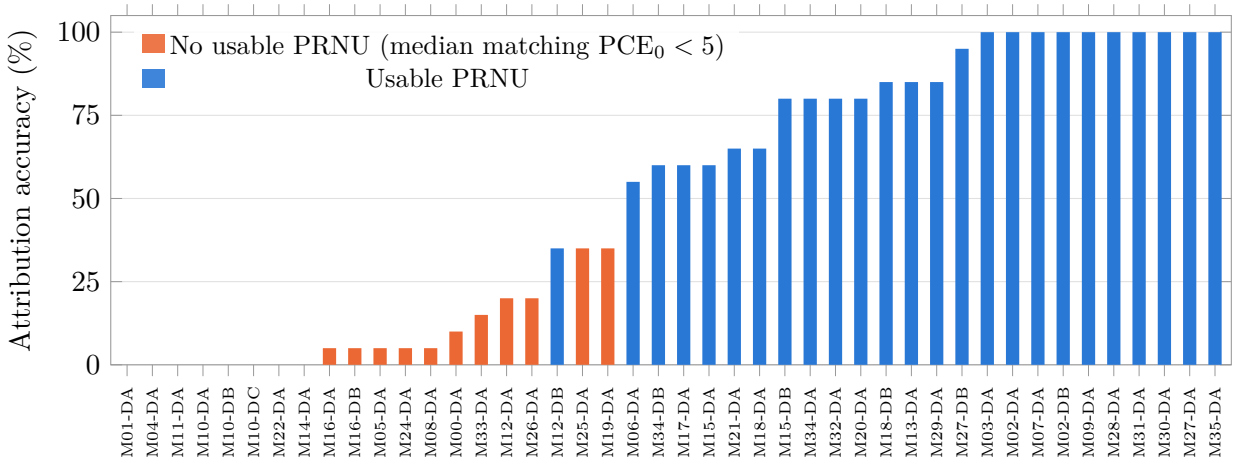


Figure 3: Per-device attribution on Video-ACID (20 sample videos per device), sorted. Devices are ACID model ids (e.g. M10 = Google Pixel 1, M00 = iPhone 8 Plus, M35 = Yi 4K action camera).

DSLRs (Canon EOS SL1/T6i), action cameras (GoPro, Yi 4K) and most Samsung phones reach 95–100% with matching PCE in the hundreds to thousands, whereas 19 devices (Google Pixel 1 and 2, iPhone 8 Plus, LG Q6, Asus Zenfone 3, several compact cameras and camcorders) have a median matching PCE below 5, most likely because of electronic image stabilisation. They are flagged as *PRNU-weak* for the analysis; the splits are unchanged.

8 Manipulated class (Step 5)

8.1 Attack design

The targets are the 400 natural native VISION videos (up to 12 per device). The first 30 s of each are decoded once in sensor orientation, cropped like the analysis frames, attacked on the luma channel only, and re-encoded with libx264 (CRF 20, one I-frame per second, about 30 I-frames per clip, matching the organic samples). Table 6 lists the 11 variants per target.

Table 6: Variants generated per target video. α is in units of a 1% PRNU: attacker fingerprints are rescaled to standard deviation 0.01.

Variant	Operation	Label	Claimed camera
`reencode`	same encoder, no attack (control)	organic	own
`removal`, $\alpha \in \{0.25, 0.5, 1\}$	$F' = F - \alpha F \hat{K}_{\text{own}}$	manipulated	own
`removal_opt`	white-box: α tuned per clip until NCC with the analyst's reference is ≈ 0	manipulated	own
`injection`, $\alpha \in \{0.25, 0.5, 1\}$	$F' = F + \alpha F \hat{K}_{\text{victim}}$ (fingerprint-copy attack [8])	manipulated	victim
`denoise`, $s \in \{1, 2, 4\}$	hqdn3d denoising at $s\times$ default strength (desynchronisation)	manipulated	own

Attacker knowledge. The attacker never uses the analyst's flat-field reference. $\hat{K}_{\text{own}}$ is estimated from the *other half* of the device's natural videos, and $\hat{K}_{\text{victim}}$ from all natural videos of a non-stabilised victim of another camera model in the same cross-validation fold (so no test camera enters training). The white-box removal is the deliberate exception: it represents the worst case for the PCE baseline. A natural-video-only calibration was tried first but gave values of α about five times too small: VISION films each device's natural videos at the same places, so two natural-video fingerprint estimates share scene residue. The white-box attacker therefore calibrates α by secant steps on the analyst's own measurement (median $\alpha^* = 0.39$, interquartile range 0.24–0.57).

Control. The `reencode` clips pass through exactly the same encoder as the attacks and are labelled organic, so encoder traces alone cannot separate the classes.

8.2 Verification

Table 7 shows PCE_0 against the analyst's references for two test videos.

Table 7: PCE_0 of attacked clips against the analyst's references (own camera / injected victim).

	D01 (victim D30)		D08 (victim D28)	
Variant	own	victim	own	victim
Original native	563	1	2,771	0
`reencode`	195	0	1,381	1
`removal` 0.25 / 0.5 / 1	12 / −199 / −1,296	–	644 / 3 / −4,765	–
`removal_opt`	0.1 (α^*=0.30)	–	0.0 (α^*=0.52)	–
`injection` 0.25 / 0.5 / 1	146 / 91 / 18	49 / 315 / 919	1,343 / 1,185 / 582	39 / 275 / 1,357
`denoise` 1 / 2 / 4	103 / 75 / 38	–	1,124 / 918 / 454	–

All variants behave as intended: injection makes the victim match while the true camera remains detectable; removal over-subtracts quickly and leaves a strongly *negative* PCE, a trace that a classifier can learn; the white-box removal lands at PCE≈ 0, the hardest case for the baseline. Output: 4,400 clips and 131,879 frames with no failures.

9 Synthetic class (Step 6)

Table 8: AI-generated clips used (after quality checks). Frames = median I-frames per clip.

Source	Generator	Native format	Frames	Train	Val	Test	Unseen test
GenVidBench	Sora	1080p, ∼17 s	18	26	17	8	–
	Kling	720p, 5–10 s	6	151	74	44	–
	CogVideo	480×480, 4 s	5	166	76	58	–
	OpenSora	512×512, 2 s	2	141	89	42	–
GenBuster	4 generators[a]	1024^2 HEVC, 5 s	6	–	–	–	4 × 150
Total				484	256	152	600

[a] CogVideoX, EasyAnimate, HunyuanVideo and LTX-Video, 150 clips each.

Generator files contain almost no I-frames (e.g. one per Kling clip), so the synthetic clips go through the same libx264 pipeline as the attacks and the control (one I-frame per second, same crop). Frame type, spacing and encoder are therefore identical across classes. Denser sampling of short clips was rejected, because frames closer in time inflate TSNCS and would create a false "synthetic" cue. Each clip claims a random camera (VISION or ACID) of a random fold, so the PRNU features can be computed; its split follows that camera. GenBuster is held out as an unseen-generator test set; its real videos are not used because they have no camera reference.

10 Feature extraction (Step 7)

Table 9: Feature groups. Per-frame values are aggregated (mean, standard deviation) per clip.

Group	Features
PRNU	per frame: NCC and PCE_0 between the residual W and $I\hat{K}$ of the claimed camera; per clip: PCE, PCE_0 and NCC of the clip fingerprint with the claimed camera, and the best match among all *other* known cameras (injection leaves a second fingerprint)
Residual	standard deviation, skewness and kurtosis of the noise residual
FFT	spectral flatness, high-frequency energy share, peak ratio, number of periodic peaks, 8-pixel block-grid energy of the residual spectrum
TSNCS	normalised correlation between residuals of consecutive I-frames (mean, std, minimum)
Noiseprint++	map std and kurtosis, FFT statistics of the map, spectral similarity to the claimed camera's reference (GPU)

Fixed-length windows. Clips differ in length (VISION about 30 I-frames, ACID about 6, synthetic 2–20). All model inputs are therefore computed from the *first five* I-frames of every clip, so clip length cannot act as a shortcut.

Noiseprint++. Noiseprint++, the pretrained extractor of TruFor [11] and successor of Noiseprint [10], runs on the GPU. In a test on 10 VISION cameras its maps did not attribute cameras: 20% top-1 with spatial similarity and 17.5% (25% at model level) with spectral similarity, against 10% chance. It was trained on RGB still images, whereas our frames are luma-only, heavily compressed video I-frames. Its output statistics are kept as a feature group, and the ablation will quantify their contribution.

Encoder confound and the controlled protocol. The Noiseprint 8-pixel grid energy is about 4 for every clip we encode with libx264 and about 0.65 for organic WhatsApp clips: our encodes start their macroblock grid at the crop origin, camera originals do not. To prevent a classifier from exploiting this, results will be reported under two protocols:

- **Realistic:** all organic samples vs manipulated and synthetic;
- **Encoder-controlled:** only libx264 clips (`reencode` controls vs attacks vs synthetic), where encoder and grid are identical across classes.

Preliminary check. On a random check set of 100 samples, the median clip PCE_0 with the claimed camera was 20.7 for organic, 61.9 for manipulated (inflated by injection) and 0.07 for synthetic clips, and the median TSNCS was 0.001, 0.003 and 0.019 respectively; on test clips TSNCS rose from about 0.002 at removal strength $\alpha = 0.25$ to 0.17 at $\alpha = 1$. These numbers only confirm that the extractors work; the full extraction over all 8,031 samples (2,539 organic, 4,000 manipulated, 1,492 synthetic) is running.

11 Models and evaluation (Steps 8–9)

- **Conventional baseline:** organic if PCE_0 with the claimed camera exceeds 60 (binary decision; it cannot tell manipulated from synthetic).
- **Classical ML:** SVM (RBF kernel, C chosen on validation) and XGBoost (depth chosen on validation, early stopping) on the clip-level features.
- **Deep temporal:** GRU and 1D-CNN on the per-frame feature sequences, pooled and joined with the clip-level features.

- **Transformer:** a two-layer Transformer encoder on the same sequences, as in the proposal.

Networks use Adam (learning rate 0.001), L2 weight decay (10^{-4}), a class-balanced cross-entropy loss and early stopping on validation macro-F1. Every configuration is run under both protocols and both schemes (fixed split and 5-fold cross-validation), with GenBuster as the unseen-generator test.

Metrics. PVA (the share of videos assigned to their true provenance category, i.e. 3-class accuracy, as defined in the proposal), macro precision, recall and F1, one-vs-rest ROC-AUC, and ROC-AUC and EER for authentic versus non-authentic (where the PCE baseline is compared, RQ3). For RQ2 the detection rate is broken down by version (native, YouTube, WhatsApp), attack type and strength α, generator, and stabilisation. The ablation removes one feature group at a time (PRNU, residual, FFT, TSNCS, Noiseprint), with special attention to TSNCS.

Status. All model and evaluation code has been tested end to end on a check set. The full runs (14 configurations) start automatically once the feature extraction finishes.

12 Deviations from the proposal

Table 10: Changes with respect to the thesis proposal and their justification.

Aspect	Proposal	Implementation and reason
Datasets	PRNU-Bench, VISION, Dresden	VISION, Video-ACID, GenVidBench, GenBuster, FloreView (pending): PRNU-Bench and Dresden are image datasets; the thesis is about video
Split	random 80:10:10	model-grouped 60/20/20 plus 5-fold CV: test cameras (and camera models) must never be seen in training, and 35 devices are too few for one small test split
Crop	512×512 suggested	480×480: two VISION devices record 480 pixels high
Classifier	Transformer	PCE baseline, SVM, XGBoost, GRU, 1D-CNN and Transformer, to answer RQ3
Noiseprint	similarity to a reference	output statistics plus spectral similarity, since attribution with Noiseprint++ is weak on our frames
Evaluation	one protocol	realistic and encoder-controlled protocols (encoder confound)

13 Technical issues and resolutions

Issue	Resolution
Only about 50 GB free disk for 280 GB of video data	ACID streamed from its archives; frames of attacks and synthetic clips stored on a second partition
GPU unavailable (NVIDIA driver not loaded)	resolved after a reboot; CUDA verified on the RTX 3080
PRNU library incompatible with NumPy 2, RGB-only estimator	minimal patch (kept as a patch file); grayscale estimator added and verified against the original
Platform copies rotated and rescaled	rotation undone and frames rescaled to native geometry; residual misalignment of rotated WhatsApp copies analysed (RQ2)
Camcorder `.MTS` files and unreadable Nokia MP4 files in ACID	`.MTS` supported; Nokia 6.1 excluded and documented
Attack strength calibration	α grid plus a white-box attacker calibrated on the analyst's measurement
Clip length and encoder differences between classes	fixed five-frame windows; shared libx264 pipeline with organic re-encode controls; encoder-controlled protocol

14 Current status and next steps

Step	Content	Status
1	Inventory and quality control	done
2	Splits and folds	done
3	I-frame extraction	done
4	PRNU fingerprints and PCE baseline	done (VISION, ACID)
5	Manipulated class	done (4,400 clips)
6	Synthetic class	done (1,492 clips)
7	Feature extraction (incl. Noiseprint++)	implemented; full run in progress
8	Models	implemented and tested; queued
9	Evaluation and ablation	implemented and tested; queued

Next steps.

1. Analyse the model results: per-class performance, comparison with the PCE baseline (RQ3), detection rate per version and attack strength (RQ2), and feature importance and ablation (RQ1).
2. Add FloreView as the cross-dataset test for organic videos (download in progress).
3. Optionally test robustness of the detectors when attacked clips are themselves shared through a social-media-like re-encoding.
4. Write the results and discussion chapters.

Points for discussion with the supervisor.

- Approval of the dataset change (video datasets instead of PRNU-Bench and Dresden) and of the model-grouped split with 5-fold cross-validation.
- Whether the encoder-controlled protocol should be the primary result.
- Whether to re-extract RGB frames to give Noiseprint++ a fair chance, at a substantial cost in time and storage.

- Scope of the FloreView cross-dataset experiment.

References

[1] D. Shullani, M. Fontani, M. Iuliani, O. Al Shaya, A. Piva, "VISION: a video and image dataset for source identification," *EURASIP Journal on Information Security*, 2017.

[2] B. Hosler, O. Mayer, B. Bayar, X. Zhao, C. Chen, J. A. Shackleford, M. C. Stamm, "A video camera model identification system using deep learning and fusion," *IEEE ICASSP*, 2019 (Video-ACID dataset).

[3] D. Baracchi, D. Shullani, M. Iuliani, A. Piva, "FloreView: an image and video dataset for forensic analysis," *IEEE Access*, 2023.

[4] "GenVidBench: A 6-million benchmark for AI-generated video detection," *AAAI*, 2026, arXiv:2501.11340.

[5] "BusterX: MLLM-powered AI-generated video forgery detection and explanation," arXiv:2505.12620, 2025 (GenBuster-200K dataset).

[6] J. Lukáš, J. Fridrich, M. Goljan, "Digital camera identification from sensor pattern noise," *IEEE Trans. Information Forensics and Security*, 2006.

[7] M. Chen, J. Fridrich, M. Goljan, J. Lukáš, "Determining image origin and integrity using sensor noise," *IEEE Trans. Information Forensics and Security*, 2008.

[8] M. Goljan, J. Fridrich, M. Chen, "Defending against fingerprint-copy attack in sensor-based camera identification," *IEEE Trans. Information Forensics and Security*, 2011.

[9] S. Mandelli, P. Bestagini, L. Verdoliva, S. Tubaro, "Facing device attribution problem for stabilized video sequences," *IEEE Trans. Information Forensics and Security*, 2020.

[10] D. Cozzolino, L. Verdoliva, "Noiseprint: a CNN-based camera model fingerprint," *IEEE Trans. Information Forensics and Security*, 2020.

[11] F. Guillaro, D. Cozzolino, A. Sud, N. Dufour, L. Verdoliva, "TruFor: leveraging all-round clues for trustworthy image forgery detection and localization," *IEEE/CVF CVPR*, 2023.

[12] L. Bondi, P. Bestagini, N. Bonettini, "prnu-python: Python porting of the PRNU extractor," Politecnico di Milano, https://github.com/polimi-ispl/prnu-python.

A VISION devices

Table 11: VISION devices used (native resolution; native flat and natural videos; YouTube and WhatsApp copies; split and fold).

Device	Brand	Model	Resolution	Stab.	Flat	Natural	YT	WA	Split	Fold
D01	Samsung	GalaxyS3Mini	1280×720	–	4	18	16	22	train	0
D02	Apple	iPhone4s	1920×1080	yes	4	9	13	13	train	3
D03	Huawei	P9	1920×1080	–	7	12	19	19	train	2
D04	LG	D290	800×480	–	7	12	19	18	val	3
D05	Apple	iPhone5c	1920×1080	yes	7	12	19	19	train	1
D06	Apple	iPhone6	1920×1080	yes	5	12	15	17	val	2
D07	Lenovo	P70A	1280×720	–	7	12	19	19	test	0
D08	Samsung	GalaxyTab3	1280×720	–	13	24	34	36	train	1

Device	Brand	Model	Resolution	Stab.	Flat	Natural	YT	WA	Split	Fold
D09	Apple	iPhone4	1280×720	–	7	12	19	19	train	0
D10	Apple	iPhone4s	1920×1080	yes	4	11	15	15	train	3
D11	Samsung	GalaxyS3	1920×1080	–	7	12	19	19	val	1
D12	Sony	XperiaZ1Compact	1920×1080	yes	7	12	19	19	test	3
D13	Apple	iPad2	1280×720	–	4	12	16	16	train	1
D14	Apple	iPhone5c	1920×1080	yes	7	12	19	19	train	1
D15	Apple	iPhone6	1920×1080	yes	7	11	18	18	val	2
D16	Huawei	P9Lite	1920×1080	–	7	12	19	19	val	2
D17	Microsoft	Lumia640LTE	1920×1080	–	4	6	10	10	train	4
D18	Apple	iPhone5c	1920×1080	yes	5	8	13	13	train	1
D19	Apple	iPhone6Plus	1920×1080	yes	7	12	19	19	test	2
D20	Apple	iPadMini	1920×1080	yes	4	12	16	16	val	0
D21	Wiko	Ridge4G	1920×1080	–	4	7	11	10	test	4
D22	Samsung	GalaxyTrendPlus	1280×720	–	4	12	16	15	val	3
D23	Asus	Zenfone2Laser	640×480	–	7	12	19	19	test	2
D24	Xiaomi	RedmiNote3	1920×1080	–	7	12	19	19	train	3
D25	OnePlus	A3000	1920×1080	yes	7	12	19	19	train	0
D26	Samsung	GalaxyS3Mini	1280×720	–	4	12	16	16	train	0
D27	Samsung	GalaxyS5	1920×1080	–	7	12	19	19	train	0
D28	Huawei	P8	1920×1080	–	7	12	19	19	train	1
D29	Apple	iPhone5	1920×1080	yes	7	12	19	19	train	4
D30	Huawei	Honor5c	1920×1080	–	7	12	19	19	train	0
D31	Samsung	GalaxyS4Mini	1920×1080	–	7	12	19	19	train	4
D32	OnePlus	A3003	1920×1080	yes	7	12	19	19	train	0
D33	Huawei	Ascend	1280×720	–	7	12	18	19	val	1
D34	Apple	iPhone5	1920×1080	yes	13	19	18	32	train	4
D35	Samsung	GalaxyTabA	1280×720	–	4	12	16	16	train	2